

An Internship Report
Of
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B-Tech (CSE)



By

Roll Number of Student:- 25SCS1003004666

Name of Student: UJJWAL KUMAR

Semester: 3rd Semester

Session: 2025-29

Organisation: Codec Technologies

CANDIDATE'S DECLARATION

I, HARSH SINGH, do hereby declare that the internship report titled “***Artificial Intelligence and Machine Learning***” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report presents the learning, practical activities and experience gained during my internship with Codec Technologies.

I further declare that the work presented in this report is prepared for academic submission and that the information relating to my internship has been compiled from the internship communication, offer letter and certificate provided by the organisation. I shall be responsible for any copyright or academic-integrity issues arising from material submitted by me.

Signature:_____

Student Name: UJJWAL KUMAR

Roll No.: 25SCS1003004666

Semester: 3rd Semester

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me throughout my internship. I am thankful for the guidance, encouragement and constructive feedback that helped me understand the practical side of Artificial Intelligence and Machine Learning.

I would like to thank Codec Technologies for providing me with the opportunity to undertake an internship as an Artificial Intelligence Intern. The internship gave me an opportunity to strengthen my understanding of AI/ML concepts and to relate academic knowledge to practical learning activities.

I am also thankful to my university and the School of Computer Science and Engineering for their continuous academic support. Finally, I express my gratitude to my parents and well-wishers for their encouragement throughout the internship.

Signature: _____

Student Name: HARSH SINGH

Roll No.: 25SCS1003004669

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3. INTERNSHIP COMPLETION CERTIFICATE



4. INTERNSHIP OFFER LETTER



INTERNSHIP OFFER LETTER

Dear Harsh Singh,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 20/07/2026 to 19/08/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

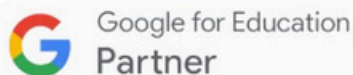
Best regards,

Sincerely,

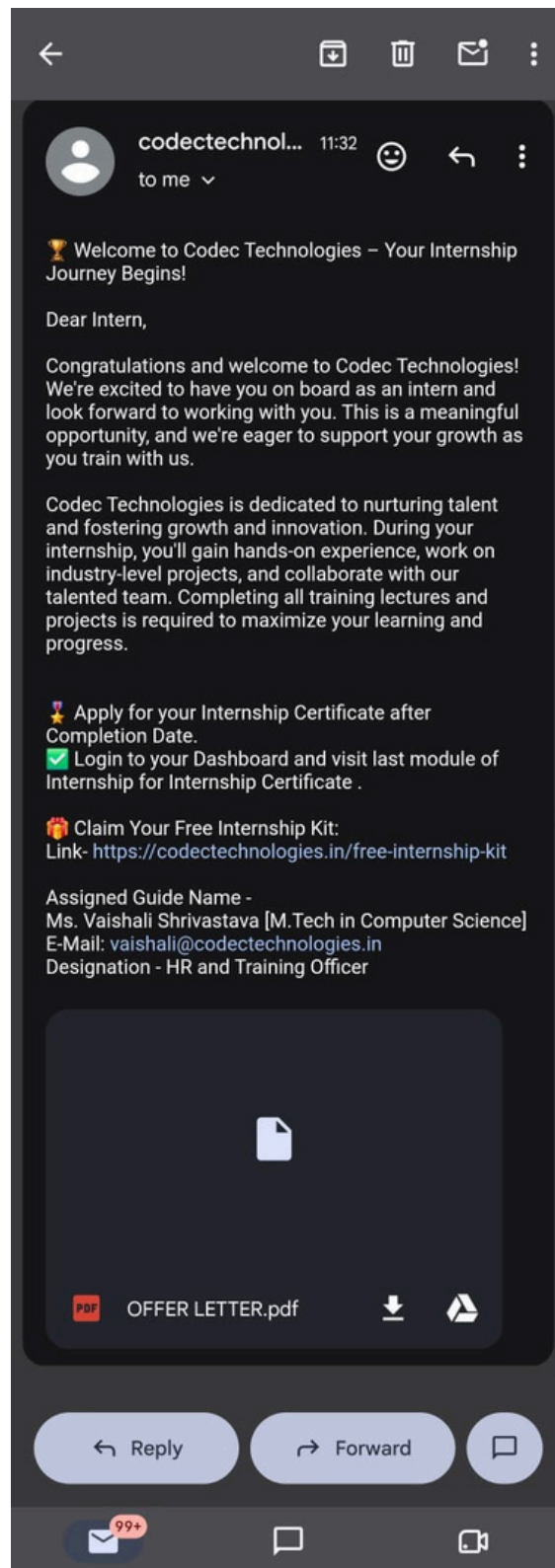
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 20/07/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



5. INTERNSHIP EMAIL / COMMUNICATION PROOF



6. PROJECT DESCRIPTION

The internship at Codec Technologies was undertaken as an Artificial Intelligence Intern from 20/07/2026 to 19/08/2026. The internship topic for this academic report is Artificial Intelligence and Machine Learning. The programme provided an opportunity to develop practical familiarity with AI/ML concepts, workflows and problem-solving approaches in an industry-oriented learning environment.

The following sections describe the internship context, objectives, scope, technologies, methodology, work performed and learning outcomes.

6.1 INTRODUCTION

Artificial Intelligence (AI) is a field of computer science concerned with developing systems that can perform tasks that normally require human-like intelligence, such as learning from data, recognising patterns, making predictions and supporting decisions. Machine Learning (ML) is a major part of AI in which computer systems learn useful patterns from data rather than relying only on explicitly written rules.

During the internship, the focus was on understanding AI/ML from a practical perspective. The learning process connected fundamental concepts with implementation-oriented activities, helping build an understanding of how AI solutions are developed, tested, evaluated and documented.

6.2 ORGANISATION PROFILE

Codec Technologies is the organisation through which the internship was offered. The offer letter describes Codec Technologies as a global platform focused on empowering learners and connecting diverse talent with career opportunities. It also states that the organisation provides IT and business consultancy services across more than 27 countries.

The offer letter records the internship designation as **Artificial Intelligence Intern**, with a Pan India, hybrid mode, a duration from 20/07/2026 to 19/08/2026, and reporting to assigned project head(s).

6.3 PROBLEM STATEMENT

A major challenge for students learning Artificial Intelligence and Machine Learning is moving from theoretical concepts to practical implementation. Understanding algorithms alone is not sufficient; an AI/ML workflow also requires suitable data preparation, implementation, evaluation, interpretation and documentation.

The internship therefore addressed the need to gain practical exposure to AI/ML-oriented problem solving and to understand how concepts studied in coursework can be applied in structured technical tasks and projects.

6.4 PROJECT OBJECTIVES

The main objectives of the internship were:

- To strengthen fundamental knowledge of Artificial Intelligence and Machine Learning.
- To understand the typical workflow used to solve AI/ML problems.
- To gain practical exposure through assignments, exercises and project-oriented activities.
- To understand data preparation, model-building concepts and evaluation at a practical level.
- To improve Python and problem-solving skills relevant to AI/ML applications.
- To develop technical documentation and communicate the work performed during the internship.

6.5 SCOPE OF THE INTERNSHIP

The scope of the internship covered practical learning in the Artificial Intelligence and Machine Learning domain. It included understanding AI/ML concepts, working through practical tasks, applying programming and analytical thinking, and documenting the learning outcomes.

The internship was completed in a hybrid/virtual learning context as specified in the offer letter. The report focuses on academic and internship learning rather than claiming deployment of an AI system in a live production environment.

6.6 TECHNOLOGIES AND TOOLS

The internship work was centred on Artificial Intelligence and Machine Learning. The principal technical areas relevant to the work are summarised below:

- Python – programming and implementation of AI/ML exercises.
- Machine Learning – understanding supervised/unsupervised learning concepts and workflows.
- Data Processing – preparing and understanding datasets before model development.
- Model Evaluation – understanding the importance of testing and evaluating model results.
- AI/ML Documentation – recording methods, observations, results and learning outcomes.

6.7 METHODOLOGY

The internship followed a practical learning methodology. The work began with understanding the problem and relevant AI/ML concepts, followed by implementation-oriented exercises and analysis. The general workflow followed during the learning activities can be represented as:

<i>Step</i>	<i>Activity</i>
1	Understand the AI/ML problem and requirements
2	Study the relevant concepts and approach
3	Prepare and inspect the required data
4	Implement the selected AI/ML approach
5	Test and evaluate the results
6	Document observations and learning

6.8 WORK PERFORMED IN AI/ML

My internship work was focused on Artificial Intelligence and Machine Learning. The practical learning activities involved understanding how an AI/ML problem is approached from data and problem definition through implementation and evaluation.

Key areas of work and learning included:

- Studying core AI and Machine Learning concepts and their practical applications.
- Working with Python-based programming exercises relevant to AI/ML.
- Understanding the role of data preprocessing and preparation in machine-learning workflows.
- Exploring the stages involved in developing and evaluating an ML solution.
- Analysing outputs and observations to understand model behaviour and results.
- Preparing technical notes and documentation for the completed internship activities.

These activities helped bridge classroom learning with practical AI/ML problem-solving and improved my understanding of the overall machine-learning workflow.

6.9 LEARNING OUTCOMES

The internship improved my understanding of Artificial Intelligence and Machine Learning and helped me approach technical problems in a more structured way. The major outcomes were:

- Better understanding of AI and ML terminology and fundamental concepts.
- Improved programming and logical problem-solving practice for AI/ML tasks.
- Understanding of the importance of data quality and preparation.
- Familiarity with the stages of an AI/ML workflow from problem definition to evaluation.
- Improved ability to document technical work and communicate results.
- Greater confidence in exploring further projects in Artificial Intelligence and Machine Learning.

6.10 CONCLUSION

The internship at Codec Technologies was a valuable learning experience in the field of Artificial Intelligence and Machine Learning. It provided practical exposure to AI/ML-oriented tasks and helped connect academic concepts with structured technical work.

The experience strengthened my understanding of AI/ML workflows, programming, data handling, evaluation and technical documentation. It also encouraged me to continue developing my skills and explore more advanced AI/ML projects in the future.

7. BIBLIOGRAPHY / REFERENCES

1. Codec Technologies, “Internship Offer Letter,” issued to Harsh Singh, Designation: Artificial Intelligence Intern, dated 20/07/2026.
2. Codec Technologies, “Certificate of Internship,” confirming completion of the one-month AICTE & ICAC approved internship programme from 20/07/2026 to 19/08/2026.
3. Internship email / communication received from Codec Technologies regarding the internship journey, certificate process and internship kit.
4. Standard academic references and documentation consulted during the internship for Artificial Intelligence and Machine Learning concepts.